

Research Proposal: Marketing Claims and Nutritional Quality in Packaged Foods

My research question asks how specific health oriented marketing claims, things like high protein, low fat, no added sugar, organic, and plant based, relate to the overall nutritional quality of breakfast cereals, snack and protein bars, yogurts, and packaged snacks, and whether that relationship changes across those categories. What draws me to this question is the gap that can open up between how a product is marketed and what its full nutritional profile actually shows. A single favorable claim on a package can shape how healthy the whole product seems, even when the rest of the nutrition information tells a different story. Researchers call this a health halo: consumers form an overly positive impression of a product because of one appealing feature, regardless of whether the broader nutritional content actually supports it (McKeon & Hallman, 2024; Franco-Arellano et al., 2020).

I'll use the Open Food Facts database, which includes product level information on labels, ingredients, nutrients, food categories, and Nutri-Score. Nutri-Score, which grades products from A to E, will be my primary outcome, and I'll also look at individual measures like sugar, sodium, saturated fat, fiber, and protein. My plan is to compare products with and without various health related claims, then use ordinal logistic regression to test whether those claims predict better or worse Nutri-Scores once differences between food categories are accounted for. I also want to know whether a claim's meaning shifts depending on the type of food it appears on: does a high protein claim tell you the same thing about a yogurt that it tells you about a snack bar? Finally, I plan to flag marketing and nutrition mismatches, cases where a product carries positive health claims but still earns a weak Nutri-Score, and check whether those mismatches hold up when I look past Nutri-Score to the individual nutrients themselves.

The project should produce a research paper, a cleaned product level dataset, reproducible analysis code, and visualizations showing how different claims relate to nutritional quality. Its value lies less in asking whether foods with health claims are healthier overall and more in asking which claims actually carry useful information, under what conditions, and when a product's real nutritional profile fails to back up the positive signal on its package. More broadly, the findings could help consumers read food marketing more critically and add to the conversation about how much weight health claims should carry in everyday food choices.

I already have most of the technical foundation this project needs. Coursework and past projects in Data Analytics and Computer Science have made me comfortable using Python and R for data cleaning, exploratory analysis, visualization, regression, and handling real world datasets full of missing or inconsistent values, skills that should transfer directly to working with Open Food Facts data. The one new statistical technique I want to pick up is ordinal logistic regression, since Nutri-Score is an ordered outcome running from A to E. Rather than collapsing products into simple healthy versus unhealthy groups, this method lets me use the full ranking and see whether certain claims push products toward better or worse categories. I also plan to extend regression methods I already know by adding interaction terms to test whether a claim's relationship to nutritional quality depends on the food category.

Beyond the statistics, I'll need to learn more about how Nutri-Score is actually calculated and where its limitations lie, so I don't end up treating it as a perfect stand-in for whether a food is healthy. I'll also need to get familiar with how Open Food Facts organizes its categories and label information so I can identify the claims I'm studying consistently across the dataset. Altogether, this is a realistic extension of skills I already have, paired with room to learn a method that fits the structure of my research question well.

The core theoretical idea behind this project is the health halo effect: the tendency for one positive characteristic of a food to color how consumers judge the product as a whole. Early work on this found that nutrient claims can pull consumer attention toward the favorable information on a package and lead

people to assume good things about aspects the claim never actually addressed (Roe et al., 1999). More recent research keeps finding that front of package claims shape perceived healthfulness and purchase intent (Franco-Arellano et al., 2020). McKeon and Hallman (2024), for instance, found that simply adding a protein focused label to cereal made people see it as healthier and more nutritious, even when the rest of its nutritional profile didn't really support that impression. Work examining products actually on store shelves has similarly found that health related marketing language doesn't reliably line up with stronger nutrition profiles (Breen et al., 2020).

This body of research matters for my project because it means I shouldn't just count health claims and ask whether products with more of them are healthier. Instead, I want to look at specific claim types, compare products within similar categories, and test whether the relationship between a claim and nutritional quality shifts depending on the kind of food it's on. My approach also differs from the experimental health halo studies in one important way: I'm not measuring consumer perception directly. Since I won't be surveying anyone, I can't claim that a given product actually caused someone to experience a health halo. What I can do is check whether the kinds of marketing cues that past studies link to consumer perception are actually backed up by the nutritional characteristics of the products currently on the market.

I'm planning to work through this project in stages, getting the data and variables solid before moving into the main analysis. From September 14 to 20, I'll finalize the food categories and health related claims I'm studying and keep reading literature on health halos, food labeling, and Nutri-Score. From September 21 to October 4, I'll pull and clean the Open Food Facts data, identify the variables I need, standardize the relevant labels, and look at how much nutrition information is missing or incomplete. From October 5 to 18, I'll run exploratory analysis to see how common each claim is, how Nutri-Scores are distributed within each food category, and how individual nutrients differ between products with and without claims.

From October 19 to November 1, I'll move into the main statistical analysis, using ordinal logistic regression and interaction terms to look at how marketing claims, food category, and Nutri-Score relate to

each other, and I'll start flagging cases where health oriented claims show up alongside weak nutritional ratings. From November 2 to 8, I'll check those results against individual nutrients (sugar, sodium, saturated fat, fiber, and protein) to make sure my conclusions don't rest entirely on Nutri-Score. From November 9 to 15, I'll put together a full first draft of the paper along with the main tables and visualizations. The rest of November will go toward revising the analysis and paper based on feedback and finalizing the reproducible code and visualizations, and during the first week of December I'll complete the final paper and put together the poster or presentation.

References

- Breen, M., James, H., Rangan, A., & Gemming, L. (2020). Prevalence of product claims and marketing buzzwords found on health food snack products does not relate to nutrient profile. *Nutrients*, *12*(5), 1513.
- Franco-Arellano, B., Vanderlee, L., Ahmed, M., Oh, A., & L'Abbé, M. R. (2020). Influence of front-of-pack labelling and regulated nutrition claims on consumers' perceptions of product healthfulness and purchase intentions: A randomized controlled trial. *Appetite*, *149*, 104629.
- McKeon, G. P., & Hallman, W. K. (2024). Front-of-package protein labels on cereal create health halos. *Foods*, *13*(8), 1139.
- Roe, B., Levy, A. S., & Derby, B. M. (1999). The impact of health claims on consumer search and product evaluation outcomes: Results from FDA experimental data. *Journal of Public Policy & Marketing*, *18*(1), 89–105.